Unit 2 Pre-Test

Scientific Notation

Standards assessed: NC.8.EE.3, NC.8.EE.4

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.EE.3 Express very large and very small quantities in scientific notation and standard form.

• *I can write numbers in scientific notation and in standard form, and recognize correct form.*

Match each vocabulary term to its definition, fill in the parts of a number in scientific notation, then write each number as directed.

D scientific notation ✓	**A.** A number written the ordinary way. *Example: 2,700 or 0.05*
B coefficient ✓	**B.** The decimal factor in a × 10^n — at least 1 and less than 10. *Example: the 8.3 in 8.3 × 10^6*
A standard form ✓	**C.** Shows a small number between 0 and 1 (move the decimal left). *Example: the −2 in 5 × 10^{-2} = 0.05*
E base ✓	**D.** A number written as a × 10^n, where 1 ≤ a < 10. *Example: 8.3 × 10^6*
F positive exponent ✓	**E.** In a power of ten, the number that is multiplied repeatedly. *Example: the 10 in 10^4 = 10 · 10 · 10 · 10*
C negative exponent ✓	**F.** Shows a large number, 10 or greater (move the decimal right). *Example: the 6 in 8.3 × 10^6 = 8,300,000*

Parts of a number in scientific notation — fill in the blanks

$$3.4 \times 10^{5}$$

coefficient	base	exponent
the 3.4	the 10	the 5

A positive exponent makes a **large** *number; a negative exponent makes a* **small** *one.*

1. *Write in scientific notation*

45,000

Answer:

answer: 4.5×10^{4}

2. *Write in standard form*

3×10^{4}

Answer:

answer: 30,000

3. *Write in scientific notation*

0.0062

Answer:

answer: 6.2×10^{-3}

4. *Write in standard form*

7.1×10^{-2}

Answer:

answer: 0.071

5. *Write in scientific notation*

580,000

Answer:

answer: 5.8×10^{5}

6. *Fix the notation*

Is 12.5×10^{6} correct? Write the correct form.

Answer:

answer: 1.25×10^{7}

SCORE 4.0 Target Content

NC.8.EE.4 Perform operations with numbers in scientific notation and compare their sizes.

• *I can multiply and divide numbers in scientific notation and tell how many times larger one is than another.*

Multiply, divide, or compare. Write each answer in correct scientific notation.

7. *Multiply*

$(3 \times 10^4)(2 \times 10^3)$

Answer:

answer: 6×10^7

8. *Divide*

$(8 \times 10^9) \div (4 \times 10^5)$

Answer:

answer: 2×10^4

9. *Compare*

A blue whale has a mass of about 1.5×10^5 kg. A dog has a mass of about 3×10^1 kg. How many times greater is the whale's mass?

Answer:

answer: 5×10^3 (5,000 times)

10. *Compare*

How many times larger is 6×10^8 than 2×10^3?

Answer:

answer: 3×10^5

11. *Divide*

$(9 \times 10^{-2}) \div (3 \times 10^4)$

Answer:

answer: 3×10^{-6}

12. *Multiply*

$(2.5 \times 10^5)(2 \times 10^{-8})$

Answer:

answer: 5×10^{-3}

SCORE 5.0 Advanced Content

NC.8.EE.4 Multiply and divide numbers in scientific notation, including negative exponents, and solve multi-step real-world problems.

• *I can multiply and divide with negative exponents, fix the form of my answer, and solve real-world problems in scientific notation.*

Solve. Show your work. Write each answer in correct scientific notation (the first number must be at least 1 and less than 10).

13. *Multiply, then fix the form*

$(4 \times 10^{-3})(6 \times 10^{-5})$

Answer:

answer: $24 \times 10^{-8} \rightarrow 2.4 \times 10^{-7}$

14. *Divide, then fix the form*

$(1.2 \times 10^{-4}) \div (4 \times 10^{-9})$

Answer:

answer: $0.3 \times 10^{5} \rightarrow 3 \times 10^{4}$

15. *Multi-step, then fix the form*

$$\frac{(7.5 \times 10^{-6})(4 \times 10^{3})}{2.5 \times 10^{-4}}$$

Answer:

answer: $12 \times 10^{1} \rightarrow 1.2 \times 10^{2}$

16. *Apply*

A red blood cell is about 8×10^{-6} meter wide. A grain of sand is about 6.4×10^{-4} meter wide. How many times wider is the grain of sand than the red blood cell? Write your answer in scientific notation.

Answer:

answer: $0.8 \times 10^{2} \rightarrow 8 \times 10^{1}$ (80 times)

17. *Apply*

One bacterium has a mass of about 2×10^{-12} gram. A sample has 6×10^{8} bacteria. What is the total mass of the sample, in grams?

Answer:

answer: $12 \times 10^{-4} \rightarrow 1.2 \times 10^{-3}$ g

18. *Apply*

A hard drive holds 2×10^{12} bytes. One song file uses about 4×10^{6} bytes. How many song files can the hard drive hold?

Answer:

answer: $0.5 \times 10^{6} \rightarrow 5 \times 10^{5}$ songs